UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education O Level

MARK SCHEME for the November 2004 question papers

5090 BIOLOGY

5090/03 Paper 3 (Practical Test), maximum mark 40

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UNIVERSITY of CAMBRIDGE
International Examinations

NOVEMBER 2004

GCE O Level

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 5090/03

BIOLOGY
Paper 3 (Practical Test)

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1 (a) (i)

Table 1.1

	for reducing sugar	for protein	for bile salts
reagent used	Benedict's solution	biuret reagents	sulphur
how treated	added reagent then heated; in waterbath ; ;	added reagent,shook/no heat ;	put reagent on surface of solution ;
result if present	yellow/orange/red (ppt) ; R:green	turns mauve/violet etc. ;	sulphur/powder sinks ;
result if not present	stays/turned blue/no reaction R: nothing happens	stays/turned blue /no reaction ; R: nothing happens	sulphur floats

[8]

Table 1.2

	test results		
urine solution	**reducing sugar**	**protein**	**bile salts**
A	stayed blue/-ve ;	turned mauve/+ve ;	given
B	turned orange/+ve	given	given
C	given	given	sulphur floated/-ve ;
D	given	stayed blue/-ve ;	sulphur sank/+ve

[4]

(ii) A (contains protein, caused by) kidney disease;
B (contains glucose, caused by) diabetes;
C healthy patient;
D (contains bile salts, caused by) liver disease; **[4]**

(b) Bile emulsifies;
fat/oil;
description of emulsification/increasing s/a of droplets;
for enzyme to 'get at';
ref. lipase;
ref hydrolysis **up to [4]**

Total : 20

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2 **(a)** Drawing marks: Body **B.[4]**

1. At least 8 cm long, clear, clean and realistic.
2. Correctly (curved and) proportioned, esp, final segment
3. Segmentation right across abdomen
4. Large gill-cover.

Appendages **A.[4]**

1. Main (thoracic) legs indicated.
2. Eye clear.
3. Antenna or base thereof.
4. Tail, with multiple components.

(b) (i) Drawing marks: **D.[3]**

1. At least 5 cm, clear, clean and realistic, with terminal cerci
2. Central and 4 lateral components shown.
3. Ciliation correctly attempted.

(ii) Magnification marks:

1. Line correctly ruled and measured.
2. Both measurements with appropriate units.
   [if cm – 0.1 accuracy]
3. Expression and calculation clearly set out.
4. Magnification and record suitable. **[4]**
   [up to 2 d.p.]

(c) (i) $C_6H_{12}O_6 + 6O_2$; = $6CO_2 + 6H_2O$;
[must balance]
or: glucose + oxygen; = carbon dioxide + water**;** **[2]**

(ii) Three from: salt water;
aeration – by pump or plants ;
<u>light</u> for photosynthesis;
food for respiration;
control of temp; (R. if ref enzymes' optimum)/pH
filter/change water + reason; **up to [3]**

Total : [20]

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